2017_053

Dec 21st

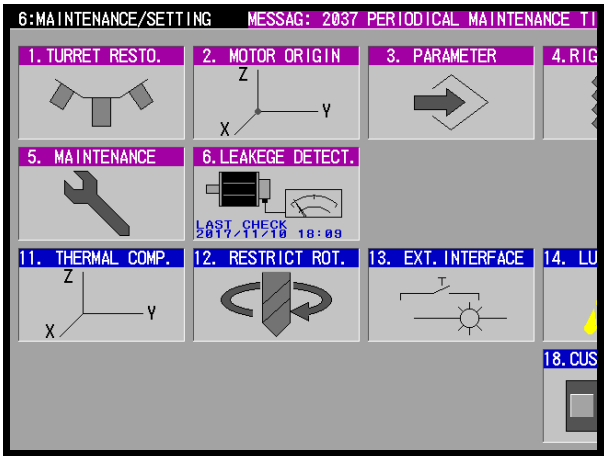
2017

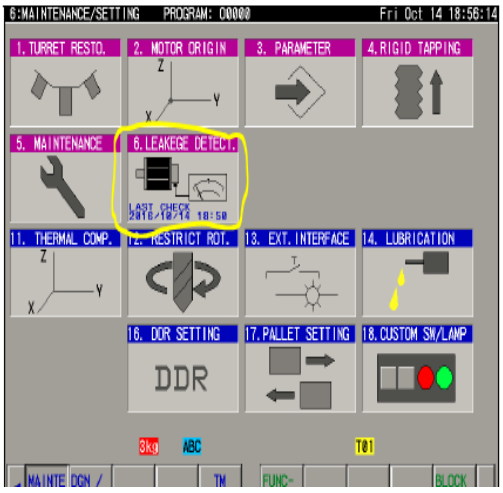
[This procedure describe about the setting of the Insulation degradation detection in RD (iFb onwards) and also generate alarm using the detection signal and judge the insulation level periodically and prevent machines breakdown]

[**The ROBODRILL a-DiB se1ies provides a dedicated screen that allows you to operate the insulation degradation detection function using the screen.**]

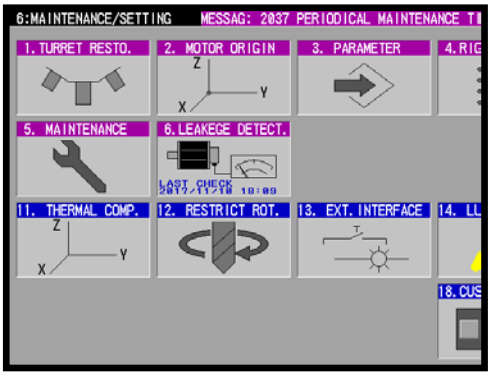
I- Explanation of Function:

1. Quick NC screen and select Maintenance / Setting





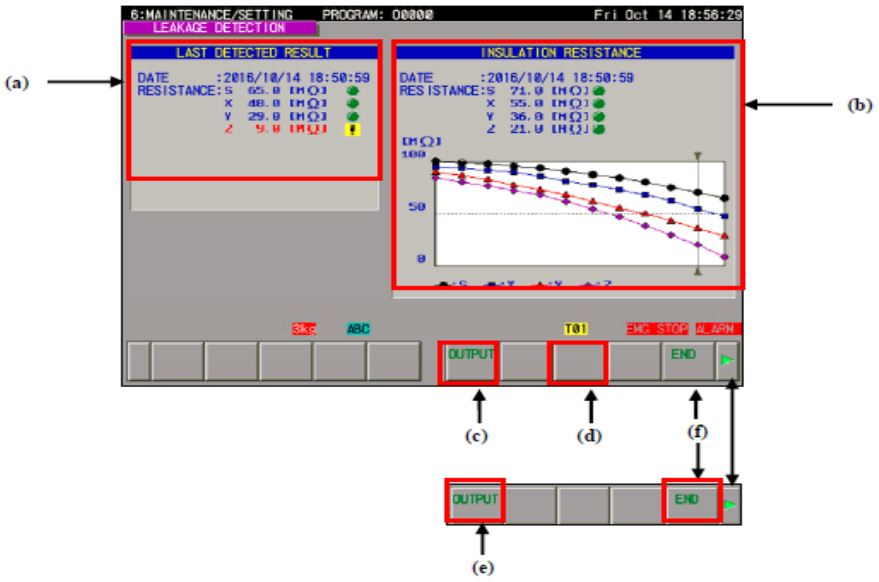
2. Select the screen Leakage detection menu :
Press “6” Input



(2) Press the <6> + <INPUT> keys. The dedicated screen for the insulation degradation detection function appears.

3.

INSULATION DEGRADATION DETECTION FUNCTION SCREEN consists of the following.



4. Explanation of functions

(a) **Final detection results**

Displays the date and time of the last execution of the insulation degradation detection function and the then measured insulation resistance value.

For details, see "47.3.2 Checking the Execution Results of the Insulation Degradation Detection Function."

(b) **Changes of insulation resistance value**

Saves the execution results of the insulation degradation detection function for up to 12 times and shows them in a graph.

By hovering the cursor over the graph, you can check the details of each data item plotted on the graph.

For details, see "47.3.2 Checking the Execution Results of the Insulation Degradation Detection Function."

(c) **[DETECT] soft key**

Displays the "EXECUTE INSULATION DEGRADATION DETECTION" window.

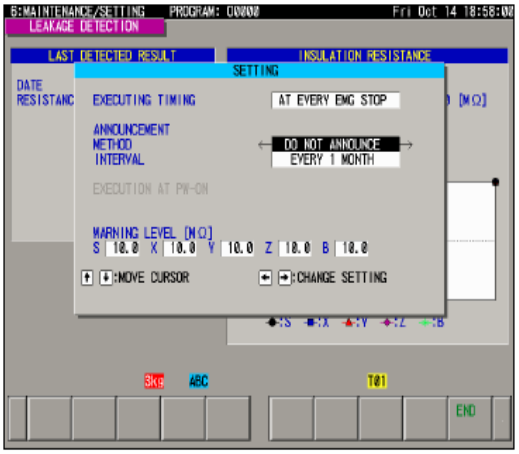
You can execute the insulation degradation detection function by operating the EMERGENCY STOP button in accordance with the guidance displayed.

For details, see "47.3.1 Executing the Insulation Degradation Detection Function."

5.

(d) **[CHANGE SETTINGS] soft key**

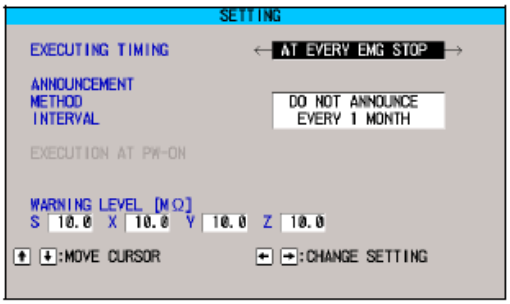
Displays the "SETTINGS" window. You can configure the following settings related to the insulatic degradation detection function.



You can set the conditions to execute the insulation degradation detection function and save detection results in the "Execute insulation degradation detection" item on the "SETTINGS" window.

6.

Select the "Execute insulation degradation detection" item using the cursor keys <↑> + <↓>, then change the setting using the cursor keys <←> + <→>.



Always execute at emergency stop (initial setting)

The insulation degradation detection function is executed and execution results are saved at the emergency stop state even on screens other than the INSULATION DEGRADATION DETECTION FUNCTIN screen.
For details on the execution of the insulation degradation detection function on non-dedicated screen, see "47.8.2 Execution on non-dedicated screens."

Execute on this screen only

The insulation degradation detection function can be executed and execution results can be saved only on the INSULATION DEGRADATION DETECTION FUNCTIN screen.
The function will not be executed on other screens, and the date and time of execution and insulation resistance value will not be saved.

7.

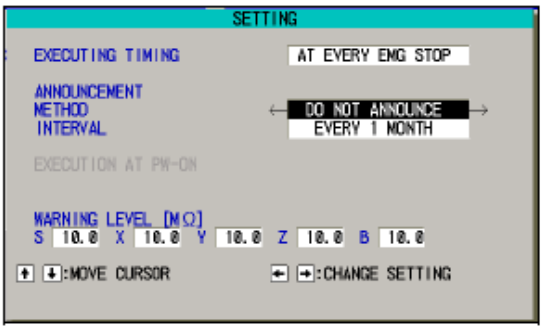
NOTIFYING EXECUTION OF INSULATION DEGRADATION DETECTION FUNCTION

8.

By displaying an alarm or operator message at a certain interval, the need to execute the insulation degradation detection function can be notified so that the function is executed at regular intervals.

9. **Settings:**

The conditions to display an alarm or operator message can be set by changing the "Notification method" and "Notification interval" items under the "Notification of execution of detection" item on the "SETTINGS" window.
Select the "Notification method" or "Notification interval" item under "Notification of execution of detection" using the cursor keys <↑> + <↓>, then change the setting using the cursor keys <←> + <→>.



10.

Notification method

Select the method to notify the execution of the insulation degradation detection function.

Not to notify (initial setting)
The execution of the insulation degradation detection function will not be notified with an alarm or operator message.

Notify with alarm
Notifies the time of executing the insulation degradation function with an alarm.

Notify with message
Notifies the time of executing the insulation degradation function with an operator message.

11.

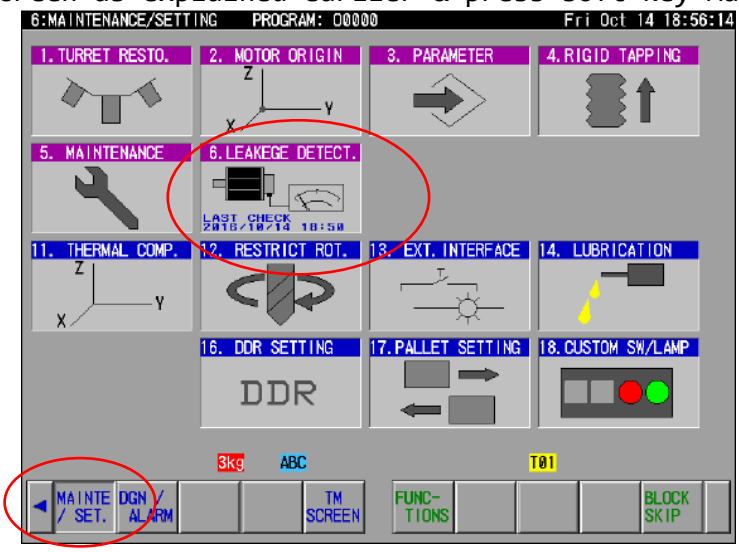
Notification interval

Select the interval to display an alarm or operator message.
You can select the interval from "every one month" (initial setting), "every three months," "every six months," and "every one year."

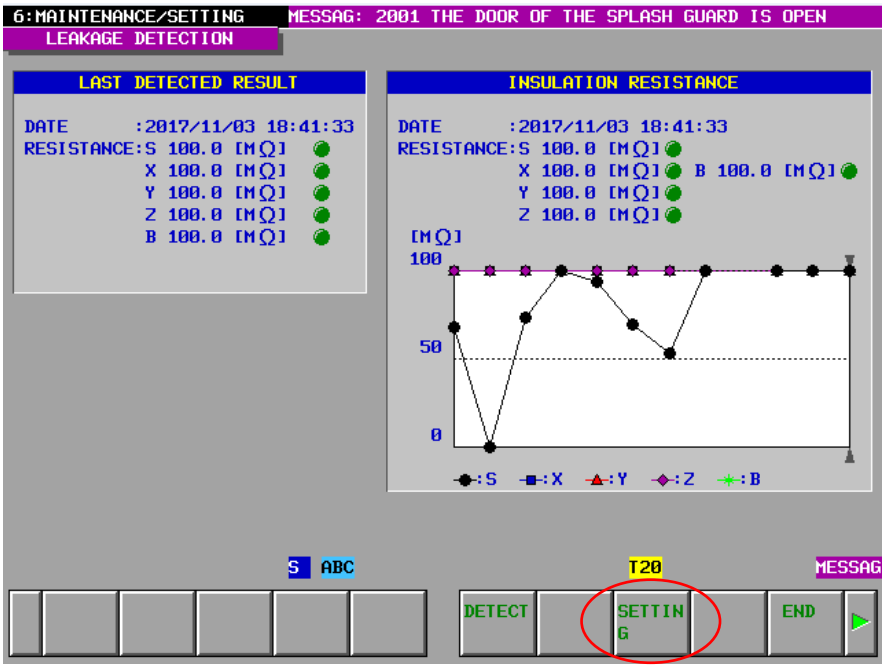
NOTE
- Here, one month is considered to have 730 hours.

II- Setting the Insulation detection values and interface with customer PMC

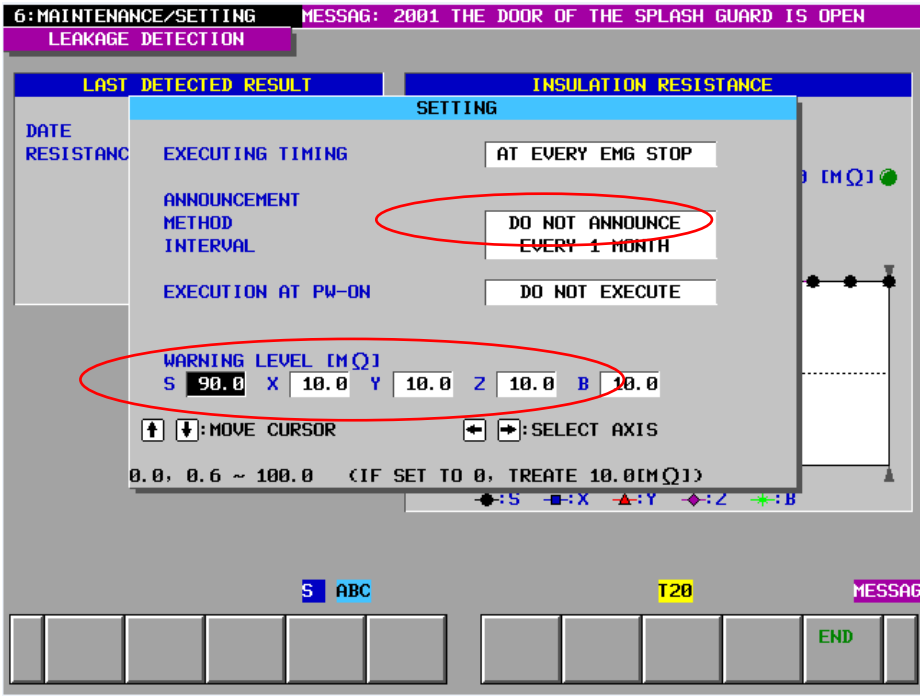
12. Go to Quick NC screen as explained earlier & press soft key Maint/Set button.



13. Type 6 & press input button on MOP, below screen appears. Press setting button on Soft key.



14. Need to do setting as required as shown in below photo.(Factory value is 10 M Ohms (you can change it to 90 M ohms or less).



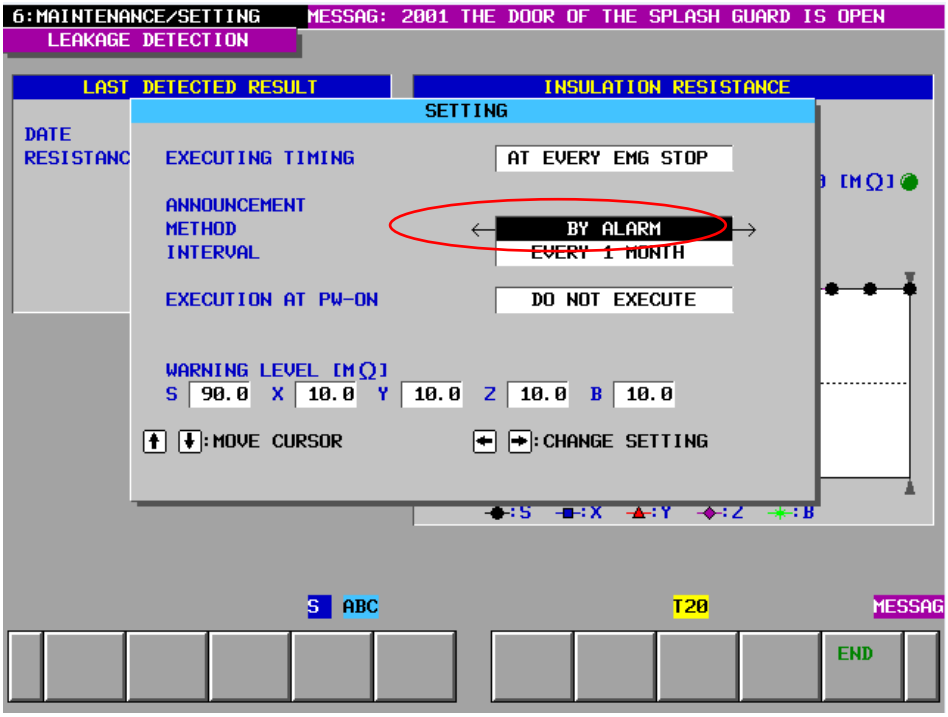
FANUC Servo motor manual information on insulation resistance:

MOTOR INSULATION RESISTANCE MEASUREMENT

Measure an insulation resistance between each winding and motor frame using an insulation resistance meter (500 VDC). Judge the measurements according to the following table. Make an insulation resistance measurement on a single motor unit after detaching cords such as a power line.

Insulation resistance	Judgment
100 MΩ or higher	Acceptable
10 to 100 MΩ	The winding has begun deteriorating. There is no problem with the performance at present. Be sure to perform periodic inspection.
1 to 10 MΩ	The winding has considerably deteriorated. Special care is in need. Be sure to perform periodic inspection.
Lower than 1 MΩ	Unacceptable. Replace the motor.

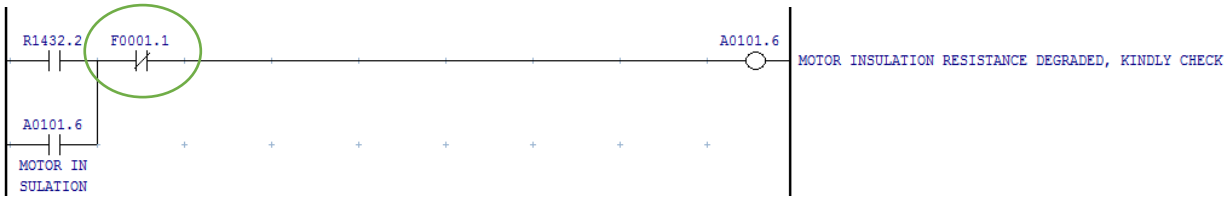
15. Set the announcement method as “BY ALARM” to generate alarm.



Above procedure is completes the setting of the insulation degradation setting. But this will not generate the alarm as soon as a detect the degradation of the insulation value. Therefore, if you want to detect the degradation, create an alarm, and want to stop the machine before it fails. You can follow below procedure to create the alarm in CUSTOM PMC program.

16. In PMC2 (Custom) ladder, need to modify as shown below to generate alarm.

Ladder screen:

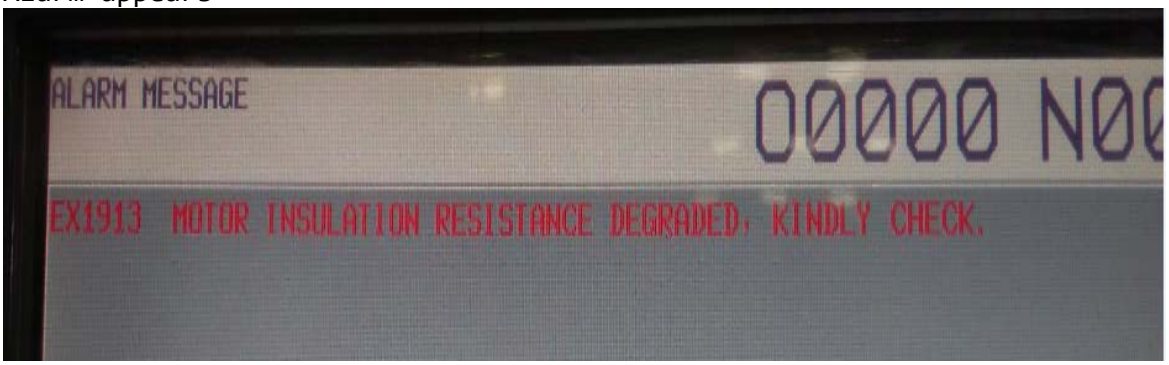


Message Screen:

815	A101.6	1913MOTOR INSULATION RESISTANCE DEGRADED, KINDLY CHECK

17. For checking purpose forced R1432.2 in PMC status screen.

Alarm appears



End of procedure-----

"For more details please refer operator manual (Alpha DiB) B85454EN-01"